

PRAMANEX

FORGE

CMC DOCUMENTATION & TRACEABILITY OS

Enterprise SaaS Master Blueprint - product, workflow, agentic automation, website, data, tests and Gemini build specification

Core product logic Controlled CMC source -> extract exact fields -> preserve raw values -> normalize carefully -> trace every field to source -> find missing/conflicting/suspicious data -> compare batches -> prepare evidence-backed draft -> human correction -> recheck -> versioned controlled export.

SOURCE	EXTRACT	NORMALIZE	TRACE	CHECK	COMPARE	DRAFT	HUMAN REVIEW	RECHECK	EXPORT
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Truth boundary: FORGE is a portfolio/product specification unless executable evidence proves more. It does not autonomously submit dossiers, release batches or establish eCTD/GxP validation.

0. Source status and architecture lock

Element	Status	Build rule
F01-F14	Canonical FORGE feature set	Preserve IDs and meaning; no silent renumbering.
L01-L12	Shared implementation alignment	Reuse across features; the original FORGE source did not itself prove a complete 12-layer implementation.
F-P01-F-P05	Proposed FORGE enhancements	Keep as child enhancements; each needs artifact + test + demo + limitation.
S-P01-S-P06	Shared hardening backlog	Apply relevant controls to intake/review/export/runtime.
Website / SaaS / agents	Proposed implementation design	Build around source-derived behavior; do not upgrade proposal into a claim of deployment.

1. Product mandate and intended users

FORGE owns CMC source-to-field/draft work: controlled source truth, structured checking, traceable draft preparation and reviewer-controlled output.

- RA / CMC reviewer - inspect source-linked technical content, gaps and conflicts.
- RegOps / submission contributor - consume approved structured draft/export packages.
- Quality / validation reviewer - inspect expected/actual tests, corrections, version history and limitations.
- Technical/admin user - manage authorized intake, workflow health, schemas, permissions and recovery.

2. Canonical FORGE features - F01 to F14

ID	Feature	Behavior	Layers
F01	CMC source intake	Accept controlled non-confidential or synthetic CMC documents/data; hold wrong-purpose or unsupported input.	L01-L04, L06
F02	Field extraction	Convert chemical/product parameters into typed candidate records with source coordinates.	L05, L09
F03	Unit / format normalization	Preserve raw value and record approved conversion; never silently alter meaning.	L04-L06, L10
F04	Source-span traceability	Every field links to exact source version and location; missing link blocks accepted output.	L04, L07, L11
F05	Completeness check	Required missing information remains missing; no invented placeholder as fact.	L05-L06, L10
F06	Wrong-value anomaly	Flag suspicious/inconsistent values for review; do not autonomously declare product/batch failure.	L05-L06, L10
F07	Cross-document consistency	Compare controlled sources and preserve disagreements rather than silently merging them.	L04-L07, L10
F08	Batch-uniformity map	Compare defined batch parameters, units and limits; variation is not automatically failure.	L05-L07, L11
F09	Submission-format preparation	Prepare structured downstream CMC content as a draft; no autonomous	L05-L06, L09-L10

Source-derived FORGE core + clearly marked proposed enterprise implementation

ID	Feature	Behavior	Layers
		submission/eCTD-conformance claim.	
F10	Human correction	Authorized reviewer accepts, corrects or rejects extracted fields with reasons; stale approval fails.	L02, L08, L10
F11	Decision / audit record	Preserve reviewer, time, evidence version, decision, original and corrected values.	L02, L04, L11
F12	Version / change record	Preserve before/after lineage for sources and fields; changed source triggers affected-field review.	L04, L08, L11
F13	Validation report	Report executed extraction/checking results, denominator, errors, exceptions and limitations.	L10-L11
F14	Controlled export	Produce reproducible source-linked evidence package; exclude unapproved/unsupported claims.	L02, L10-L12

3. Proposed FORGE additions - F-P01 to F-P05

ID	Enhancement	Behavior	Acceptance example
F-P01	Field dependency + source-change map	Link extracted source fields to downstream draft tables/sections. A changed source marks only dependent fields for re-review.	Change one synthetic assay value; affected fields become stale while original lineage remains.
F-P02	Unit + value conflict workbench	Show raw values, units, conversion rule, competing sources and unresolved conflict side by side.	mg vs g / percent vs fraction ambiguity cannot auto-accept.
F-P03	Source-backed drafting contract	Every material draft statement has source support or explicit MISSING / NEEDS REVIEW.	Remove a source parameter; draft retains the gap rather than inventing replacement.
F-P04	Reviewer correction replay	Preserve who changed a field, why and against which source; rerun checks and invalidate stale approval.	Approve v1, edit field/source, prove v2 needs new decision.
F-P05	Batch comparison with interpretation limits	Compare batches only with explicit method, unit and limit context; flag missing comparability.	Different test methods block simplistic numeric interpretation.

4. Shared 12 layers - FORGE mapping

Layer	Name	FORGE responsibility
L01	Workspace & tenant isolation	Separate organizations/projects and all CMC/evidence objects.
L02	Identity, roles & approval	Control author, reviewer, approver and export permissions.
L03	Authorized source & intake	Admit only permitted source types/purposes and safe inputs.
L04	Storage, metadata & versioning	Preserve original files, hashes, field revisions, predecessor links and decision history.
L05	Domain parsing & normalization	Parse tables/text; type fields; normalize formats

Layer	Name	FORGE responsibility
		while preserving raw values/coordinates.
L06	Source / policy / context governance	Apply document purpose, product context, units, methods, requirements and permitted transformations.
L07	Evidence graph, retrieval & analytics	Connect source span -> field -> conflict -> batch -> draft statement -> decision.
L08	Workflow state & orchestration	Move records through extract/check/review/recheck/export and manage waits/retries/stale states.
L09	Bounded multi-agent / model routing	Route extraction, consistency, drafting or comparison tasks to typed helpers.
L10	Safety, security & human decision	Block unsupported/missing/ambiguous output; preserve human reviewer control.
L11	Evaluation, audit, reporting & feedback	Run fixed fixtures, trace coverage, correction/recheck and historical replay.
L12	Integration, operations & scale	APIs, notifications, job workers, exports, backups and later RIM/QMS connections.

5. Full feature + layer + proposal linkage

Stage	Core	Proposal	Layers	Outcome
1. Controlled intake	F01	-	L01-L04, L06	Authorized source/version enters or is held.
2. Extract truth	F02-F04	F-P01	L04-L07, L09-L11	Typed candidate fields with exact source coordinates.
3. Normalize safely	F03	F-P02	L04-L06, L10	Raw value retained; conversions and ambiguities visible.
4. Check completeness	F05-F07	F-P02/F-P03	L04-L07, L10-L11	Missing/conflicting/suspicious information stays explicit.
5. Compare batches	F08	F-P05	L05-L07, L11	Comparable parameters mapped with method/unit/limit context.
6. Prepare draft	F09	F-P01/F-P03	L05-L10	Structured source-backed draft with gaps visible.
7. Human correct	F10-F11	F-P04	L02, L04, L08, L10-L11	Exact revision accepted/corrected/rejected with rationale.
8. Source changes	F12	F-P01/F-P04	L04, L08, L11	Only dependent work reopens; old decision remains historical.
9. Validate/export	F13-F14	F-P03/F-P05	L02, L10-L12	Measured results + controlled evidence package.

6. End-to-end operational workflow

INTAKE	PARSE	EXTRACT	TRACE	NORMALIZE	GAP/ CONFLICT	BATCH CHECK	DRAFT	HUMAN	RECHECK	EXPORT
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4A. Enterprise parsing + routing resilience enhancements

These controls extend the existing L05 and L09 responsibilities without renumbering or changing F01-F14 or L01-L12. They are implementation targets and remain subject to executed validation evidence.

L05 enhancement - Vision-LLM & Tabular Extraction Pipeline

Capability	Required behavior	Control boundary
Complex multimodal route	Certificates of Analysis (CoAs), raw-material test graphs and complex nested specification tables may route to a Vision-LLM / document-AI extraction path when deterministic text/table parsing is insufficient.	The original file/image, page and coordinates remain authoritative; model output is a candidate representation, not accepted CMC truth.
Typed JSON conversion	Convert visual/table observations into the approved typed JSON schema: source/version, page/region, field, raw value/text, unit, method, table path, candidate value and extraction state.	Low-confidence, unreadable or ambiguous cells remain NEEDS_REVIEW / HOLD. Never infer an absent value.
Evidence preservation	Retain original bytes, rendered page evidence, coordinates and parser/model identity alongside extracted JSON.	A JSON object cannot replace source provenance. Re-review is required after source/model/schema change.

Multimodal truth rule The design targets faithful structured extraction, but does not claim "lossless" conversion until validated against representative CoA, graph, handwriting and nested-table fixtures. Original evidence is always retained.

L09 enhancement - Deterministic Fallback & Dead-Letter Queue (DLQ)

Level	Protocol	FORGE behavior
Level 1 - Retry	3 automatic retries with exponential backoff for eligible provider/API failures.	Retry remains bounded and idempotent; repeated failure never becomes successful extraction/draft state.
Level 2 - Queue	Dead-Letter Queue for non-critical drafting, formatting and other asynchronous AI-assisted tasks.	Persist run ID, exact revision, payload reference, failure reason and retry count; recover/process after service restoration.
Level 3 - Deterministic bypass	Mandatory CMC verification continues in rule-based/manual mode when AI is unavailable.	Users retain source/version access, deterministic completeness/trace/unit checks and human review so verification work is not blocked by LLM outage.

Failure honesty AI outage must surface as provider unavailable / queued / manual-mode state. FORGE must not silently substitute a model, fabricate a draft, or mark verification complete.

- 1. User creates a CMC workspace and selects a bounded task/purpose.
- 2. Authorized synthetic/public-safe CMC documents are uploaded or connected; F01 records identity, version and permission.
- 3. Parser preserves source coordinates; F02 extracts typed fields while F04 attaches exact source locations.
- 4. F03 records raw values and any approved normalization; ambiguous conversions remain unresolved.
- 5. F05-F07 expose missing fields, suspicious values and conflicts across documents.
- 6. F08 compares defined batch parameters only where methods/units/limits are comparable.
- 7. F09 prepares structured draft sections; F-P03 requires source support or an explicit gap state.
- 8. F10 routes exact candidate revision to a named reviewer. Correction creates a new revision, not an overwrite.
- 9. F12 + F-P01 mark only dependent fields/draft sections stale when a source changes.
- 10. F13 reports extraction/checking results and limitations. F14 exports only approved source-linked content.

Golden demo fixture Use three synthetic CMC sources containing one assay value, one unit mismatch, one cross-document conflict and one missing parameter. Show extraction -> conflict/gap -> reviewer correction -> recheck -> source-linked export.

7. Website user journey

Journey step	User sees	System action
1. Sign in / workspace	Product, dosage form, task purpose, source set.	Apply tenant/role/purpose policy.
2. Source intake	Document cards with version, type, parse state and source health.	Hash/store/parse; reject unsupported input.
3. Extraction workbench	Field table: raw value, normalized value, unit, source location, confidence/status.	Create typed candidates; no silent acceptance.
4. Gap & conflict queue	Missing fields, wrong-value flags, cross-document conflicts.	Route unresolved items to human review.
5. Batch comparison	Batch x parameter matrix with method/unit/limit context.	Show comparability status and exceptions.
6. Draft composer	Structured CMC sections with statement-to-source links.	Block unsupported statements; mark NEEDS REVIEW.
7. Review queue	Side-by-side source span, extracted field, draft usage and reviewer actions.	Bind decision to exact revision; revalidate after correction.
8. Change impact	Source v2 shows dependent fields/sections that became stale.	Reopen only affected work.
9. Validation & export	Metrics, exception list, approved evidence package.	Generate controlled manifest/export.

10. Enterprise SaaS website - experience architecture

Design objective FORGE should feel like a regulated enterprise workbench, not a marketing mock-up. Every screen should help a real user understand evidence, status, ownership, risk and the next permitted action.

Primary user roles

Role	Primary job	Default authority
CMC Analyst / Author	Prepare and reconcile source-linked technical content.	Create candidate work; no final release.
CMC / RA Reviewer	Review conflicts, corrections and draft evidence.	Approve/revise/request evidence/reject exact revisions.
QA / Validation Reviewer	Inspect tests, limits, audit history and controlled	Review evidence; does not gain business approval

Role	Primary job	Default authority
	export.	automatically.
System Admin	Configure workspace, roles, allowed sources and integrations.	Administrative control; no automatic scientific authority.

Authenticated product navigation

Route / page	Purpose	Main objects / actions
/app	Command Center	Work queues, source health, conflicts, stale items, review due, export status.
/sources	CMC Sources	Upload/connect sources; version/parse/permission state.
/extraction	Extraction Workbench	Typed fields, raw/normalized values, coordinates, filters, bulk review.
/gaps-conflicts	Gap & Conflict Queue	Missing, inconsistent, suspicious and unresolved items.
/batches	Batch Comparison	Parameter matrix, units/methods/limits, comparability warnings.
/drafts	CMC Draft Workspace	Section tree, evidence-backed statements, missing/needs-review flags.
/reviews	Review Queue	Source span vs field vs draft usage; approve/correct/reject/request evidence.
/changes	Source Change Impact	v1/v2 diff + dependent field/section stale map.
/validation	Validation Report	Fixture/test outcomes, errors, denominators, limitations.
/exports	Controlled Exports	Approved source manifest, decision history, release=false by default.
/audit	Replay & Audit	Reconstruct historical source -> field -> correction -> draft -> decision.
/settings	Organisation / Settings	Roles, source policy, schema versions, notification and integration settings.

Command center / dashboard

Card / panel	What it answers	Interaction
Open work	How many extraction/draft items need action?	Open filtered queue.
Evidence gaps	Where are required data missing?	Jump to gap list and source context.
Conflicts	Which fields disagree across documents?	Open side-by-side workbench.
Stale after change	What became invalid after source v2?	Open dependency map.
Review due	Which exact revisions await human decision?	Open review queue.
Batch exceptions	Which comparisons are not safe to interpret?	Open batch matrix.
Trace coverage	How much accepted content has exact source linkage?	Open traceability report.
Export readiness	What blocks a controlled package?	Show blockers and owners.

Visual language

- Desktop-first enterprise shell: persistent left rail, top context bar, global search, workspace switcher, notification center, user/role menu.
- Color system: deep navy foundation; restrained cyan/blue for active evidence; amber for needs-review; red only for blocked/critical; green only for verified/approved states.
- Typography: modern grotesk/sans family; compact 12-14 px operational text, 16-20 px section titles, 28-40 px hero statements. Use clear hierarchy, not decorative typography.
- 3D/motion: use subtle depth in the public landing page and architecture visualization only. Inside regulated work screens, motion is functional: state transition, trace highlight, dependency pulse and review confirmation.
- Never hide uncertainty behind animation. State, source version, evidence completeness, human authority and failure reason remain textually visible.

Public landing page

Section	Content	Motion / visual
Hero	FORGE identity + one-sentence problem/outcome + controlled demo CTA.	Slow 3D evidence-network / document-stack motion; no distracting particles.
Problem	CMC teams lose time reconciling values, units, versions, missing fields and downstream draft impact across controlled sources.	Before/after workflow strip.
How it works	Controlled source -> exact field -> checks -> source-backed draft -> human correction -> recheck -> controlled export	Scroll-linked stage illumination.
Evidence & human control	Exact source/version links, human gate, failure states, audit reconstruction.	Interactive evidence-chain hover.
Product screens	Command center, workbench, review, audit/export screenshots.	Subtle parallax cards.
Architecture	Frontend -> API -> orchestration -> data/evidence -> evaluation -> human gate.	Layered 3D stack, reduced-motion fallback.
Proof / limitations	Executed tests, demo scope, known limits, synthetic/public-safe data label.	Static; evidence-first.
CTA	Open controlled demo / view case study / GitHub evidence as actually available.	Minimal hover only.

11. Frontend, backend, database and infrastructure

Reference implementation stack

Plane	Recommended implementation	Boundary
Frontend	Next.js / React + TypeScript + accessible component system; private operational UI may use Streamlit for a bounded prototype.	Do not confuse a public portfolio shell with the controlled review application.
API	Python + FastAPI + Pydantic / JSON Schema.	Typed input/output; reject malformed or unauthorized work before model calls.
Auth / tenancy	Supabase Auth + PostgreSQL row-level security or equivalent.	Tenant, role and purpose checks at relational, object, retrieval and export paths.
Database	PostgreSQL + versioned relational records; pgvector only where retrieval adds value.	Relational truth first; vector retrieval never replaces source/version evidence.
Object storage	Versioned controlled documents, manifests and exported evidence packages.	Original bytes retained; safe content types and access policies.

Plane	Recommended implementation	Boundary
Automation	n8n parent workflow + named bounded specialist subflows.	Deterministic routing, waits, retries and audit calls owned by one parent orchestrator.
AI runtime	Pinned model/configuration gateway with explicit fallback and recorded model identity.	No silent model substitution; task-specific evaluation before release.
Evaluation	pytest + fixed fixtures + structured outcome logs; optional RAG evaluation where appropriate.	Report denominators, failures and limitations.
Observability	Structured logs + trace IDs + workflow metrics + product analytics.	Never send regulated/confidential payloads to analytics by default.
Deployment	Containerized API/worker; managed PostgreSQL/storage; scheduler/worker runs independently of browser.	Live means deployed endpoint actually tested; production requires additional security/recovery/validation work.

Core data objects

Object	Minimum fields / relationships
workspaces	tenant_id, product/context, purpose, schema/workflow versions, status
source_documents	source_id, file/object URI, type, permission, version, predecessor, hash, timestamps, parse status
source_spans	source_id/version, page/section/table/cell coordinates, raw text/value
extracted_fields	field_id, field type, raw value, normalized value, unit, source_span_id, candidate revision, status
normalizations	field_id, rule/version, raw/converted value, rationale, reviewer state
conflict_records	field ids/source spans, conflict type, unresolved/resolved state, rationale
batch_parameters	batch_id, parameter, method, unit, value, limit, comparability state
draft_sections	section_id/version, statements, field dependencies, evidence links, status
field_decisions	candidate revision, reviewer, action, reason, timestamp, residual issue
validation_runs	fixture hash, expected/actual, denominator, failures, environment, commit
exports	manifest, included revisions/evidence, reviewer, redactions, release flag, created_at
audit_events	ordered event id, actor, object/revision, action, correlation_id, timestamp

API surface - proposed

Method / route	Purpose	Safety / state rule
POST /sources	Register/upload controlled source.	Purpose/type/tenant gate before parse.
POST /sources/{id}/extract	Run typed extraction.	Idempotent per source revision + extractor config.
GET /fields	Query extracted fields/evidence.	Tenant + role + source permission filtered.
POST /fields/{id}/normalize	Propose approved conversion.	Preserve raw value; ambiguous -> HOLD.
GET /conflicts	List missing/conflicting/anomalous items.	Never auto-resolve evidence conflict.
POST /drafts	Create draft from selected approved candidates.	Unsupported statement -> NEEDS_REVIEW.
POST /reviews/{id}/decision	Approve/correct/request evidence/reject revision.	Expected-revision guard; correction creates revision.

11A. Data model / schema definition - MDM readiness

This schema rule extends the existing Core data objects section. It does not replace FORGE local identifiers or evidence relationships.

Entity family	Required MDM-ready fields	Purpose
Product	local Product_ID + optional external_uuid + external_system + mapping_status	Future cross-module/product-master correlation without changing existing local primary key.
Batch	local Batch_ID + optional external_uuid + external_system + mapping_status	Link batch evidence to an authoritative enterprise master while preserving FORGE provenance.
Facility	local Facility_ID + optional external_uuid + external_system + mapping_status	Enable future site/facility correlation across RIM/QMS/CMC systems.
User	local User_ID + optional external_uuid + external_system + mapping_status	Support future identity/master-data correlation while keeping application authorization separate.

Mandatory MDM rule

All local entities (Product_ID, Batch_ID, Facility_ID, User_ID) must accept external UUIDs for future Master Data Management (MDM) readiness.

- External UUIDs are correlation keys, not proof of authority. Source authority still comes from configured enterprise master/source governance.
- A local record may exist before an external UUID is known; mapping_status must explicitly represent UNMAPPED, MATCHED, CONFLICT or RETIRED.
- Do not overwrite local historical IDs when an external mapping changes. Store mapping history and effective timestamps.
- Cross-module joins must preserve tenant, purpose and authorization controls before exposing correlated data.

Architecture outcome FORGE becomes MDM-ready without becoming an MDM system itself: local operational truth remains intact while future PRAMANEX/RIM/QMS integrations can correlate stable external identities.

Method / route	Purpose	Safety / state rule
POST /sources/{id}/supersede	Register source v2 and compute dependency impact.	Mark only dependent fields/sections stale.
POST /validation/run	Execute fixed fixture suite.	Record environment, denominator and failures.
POST /exports	Generate controlled evidence package.	Approved supported revisions only; manifest retained.

Bounded agent / specialist model

Specialist	Job	Must not do
Source / parser specialist	Classify supported document structure; preserve coordinates.	Invent fields or discard original source.
Field extraction specialist	Extract typed candidate CMC parameters.	Turn low-confidence candidate into accepted fact.
Consistency specialist	Compare sources, values and completeness.	Silently choose one conflicting source.
Batch comparison specialist	Align comparable parameters with method/unit/limit context.	Declare product/batch disposition.
Drafting specialist	Compose bounded source-backed draft statements.	Create unsupported regulatory claims or submit dossier.
Evidence verifier	Check every accepted field/statement against source revision.	Override human decision.

Agentic design rule Use agents only for bounded candidate work. Deterministic checks, permissions, state transitions, persistence and release gates remain code/workflow responsibilities. No decorative swarm and no autonomous regulated disposition.

12. Automation runtime and state machine

One parent workflow owns trigger, authorization, idempotency, parsing, specialist routing, validation, persistence, human waits, retries, export and terminal state. Every work item has a correlation/run ID and exact revision.

- START -> AUTHORIZED -> INPUT_VERSIONED -> PARSED -> CANDIDATE_READY -> VALIDATED -> HUMAN_REVIEW_REQUIRED -> APPROVED/REVISED/REQUEST_EVIDENCE/REJECTED -> ACTION_OPEN -> VERIFIED/CLOSED or HOLD/EXHAUSTED.
- Retries are bounded; an exhausted path is visible and owned. Never convert an error into a success state.
- Revision changes invalidate stale approvals. A human decision is bound to an exact source/input/output revision.
- Notifications use an outbox/idempotency key. A retry must not create duplicate external effects.

13. Validation, tests and measurable outcomes

Test family	Required cases	Pass meaning
Intake	valid source; unsupported type; wrong purpose; oversized/malicious path	Only permitted source enters processing.
Extraction	known fields + coordinates; altered value; missing field	Expected fields match; missing stays missing.
Units	mg/g conversion; percent/fraction ambiguity; unknown unit	Raw preserved; only approved transformation accepted.
Traceability	valid link; missing/incorrect source span	Accepted output requires exact evidence link.
Consistency	same value; true conflict; stale source	Disagreement remains visible until human disposition.

12A. Human Authority & Finalization - E-Signature Gateway

Sub-feature: **E-Signature Gateway (21 CFR Part 11 compliance-targeted)**. This control attaches to final dossier lock / controlled finalization and does not change F10-F14 semantics: the named human remains responsible for the exact revision being finalized.

Control	Required behavior	Evidence retained
Trigger	Final dossier lock and configured status transition to an approved/finalized or rejected controlled state requires a signature event.	Exact dossier/export revision, prior status, requested status and correlation/run ID.
2-factor authentication	Require Password + OTP or an approved biometric second factor before accepting the signature.	Signer identity, role, authentication event ID, factor result and UTC timestamp.
Reason for Signature	Mandatory controlled selection such as Authored, Technical Review, Regulatory Approval.	Reason code + optional bounded comment linked to exact revision.
Immutable audit evidence	Generate UTC timestamped append-oriented audit event with cryptographic hash of signed revision and signature-event metadata.	Dossier/source manifest hash, event hash, actor, reason, timestamp, predecessor/chain reference where implemented.
Post-signature change	Any material change creates a new revision and invalidates the prior finalization state for the changed content.	Prior signature remains historical; new revision requires appropriate re-review/re-signature.

Part 11 truth boundary The blueprint defines a Part 11-oriented e-signature control. A production claim of 21 CFR Part 11 compliance additionally requires validated implementation, identity/security configuration, SOPs, record-retention controls, operational procedures and documented testing in the deployment environment.

Finalization state rule

DRAFT / REVIEWED -> SIGNATURE_REQUIRED -> AUTHENTICATED -> SIGNED_REVISION_LOCKED. Failed authentication, missing reason, stale revision or hash mismatch must return HOLD / REAUTHENTICATE / RE-REVIEW rather than silently finalizing.

Human authority E-signature proves the accountable act on a bound revision; it does not allow FORGE or an AI specialist to make the scientific/regulatory decision autonomously.

Test family	Required cases	Pass meaning
Batch	same method/unit; different method; missing limit	Non-comparable cases flagged before interpretation.
Drafting	fully supported statement; removed evidence; conflicting evidence	Unsupported content blocked or NEEDS_REVIEW.
Human review	approve; correction; stale/concurrent decision	Decision bound to exact revision; stale rejected.
Change impact	source v2 modifies one field	Only dependent fields/sections reopen; history preserved.
Tenant/security	wrong tenant/role retrieval/export	No protected evidence leaks.
Recovery/export	restore then recreate package	Identifiers/hash/decision manifest reconstruct correctly.

Suggested measurable outcomes for the prototype

Metric	Definition / denominator
Field extraction match rate	matched expected extracted fields / expected fields in frozen fixtures.
Source trace coverage	accepted fields/statements with exact valid source link / accepted fields/statements.
Missing-field detection	known missing requirements correctly surfaced / expected missing items.
Conflict detection	expected source conflicts surfaced / expected conflicts.
Normalization errors	incorrect accepted conversions / conversion cases.
Reviewer correction rate	reviewer-corrected candidates / reviewed candidates; show denominator.
Stale dependency precision	correctly reopened dependent objects / all reopened objects in change fixtures.
Controlled export completeness	required approved evidence/decisions present / required export items.

Claim discipline Synthetic/public-safe results prove only the tested fixture behavior. They do not prove production validation, real-world ROI, regulator acceptance, customer adoption or enterprise security.

14. Security, privacy, recovery and enterprise hardening

- Least privilege: role and tenant checks before retrieval, review, export or model access.
- Network intake: allow-listed destinations; redirect recheck; private-network block; timeout and payload-size controls.
- Concurrency: expected-revision compare/update for all approvals and corrections.
- Secrets: server-side secret manager/environment; never expose API keys in browser bundles, repositories, screenshots or exports.
- Audit: append ordered event history, but do not claim tamper-proof without a defined threat model and tested external controls.
- Backups: restore test must reproduce at least one historical decision/package using original identifiers and hashes.
- Privacy: public demo uses labelled synthetic/public-safe data. Confidential company data needs contractual, security and environment controls outside the portfolio demo.
- Accessibility: keyboard navigation, focus states, semantic labels, contrast, reduced-motion support and non-color status cues.

15. MVP -> controlled live demo -> enterprise path

Gate	Required evidence
Specified	Feature/state/data/API/acceptance behavior documented.

Source-derived FORGE core + clearly marked proposed enterprise implementation

Gate	Required evidence
Built	Runnable component + setup instructions + actual artifact.
Tested	Executed expected-vs-actual tests; failures and limitations retained.
MVP	One useful bounded end-to-end journey works with synthetic/public-safe data and named human review.
Controlled live demo	Hosted endpoint opened and checked; worker/scheduler works without browser; access rules and operating limits recorded.
Pilot	Real organization accepts intended use, security, recovery, support, privacy and success criteria.
Production	Validated operating controls appropriate to intended regulated use. Not implied by portfolio completion.

Quality target: do not promise a permanently bug-free system. Before demo release, require zero known P0/P1 defects in the tested critical journey, no uncaught errors on tested routes, passing tenant/role negatives, tested failure states and reproducible test evidence.

16. Gemini master build prompt - paste-ready

MASTER BUILD TASK - PRAMANEX FORGE

Build a working controlled SaaS prototype for FORGE - CMC Documentation & Traceability OS. Preserve canonical F01-F14 and the shared L01-L12 mapping. Implement F-P01-F-P05 as proposed child enhancements without renumbering the core.

CORE JOURNEY

Authorized CMC source -> version/hash -> parse -> typed field extraction -> exact source-span trace -> raw/unit normalization -> completeness/anomaly/cross-document checks -> bounded batch comparison -> source-backed CMC draft -> named human correction/approval -> recheck -> source-change dependency invalidation -> validation report -> controlled evidence export.

NON-NEGOTIABLES

- Missing data stays missing; never invent a placeholder as fact.
- Preserve raw values and exact source coordinates.
- Conflicts remain visible until human disposition.
- Batch variation does not equal product failure.
- Draft is not an autonomous submission and no eCTD-conformance claim without separate evidence.
- Every human decision binds to an exact revision; stale/concurrent approval must fail.
- Source v2 reopens only dependent fields/draft sections; old history remains immutable.
- No autonomous batch release, legal clearance or regulated disposition.

UI

Create a premium enterprise shell with Command Center, Sources, Extraction Workbench, Gap & Conflict Queue, Batch Comparison, CMC Draft Workspace, Review Queue, Source Change Impact, Validation Report, Controlled Exports, Replay & Audit, Settings. Public landing page may use restrained 3D document/evidence architecture; operational pages prioritize legibility and evidence.

IMPLEMENTATION

Use typed API contracts, PostgreSQL/Supabase-style auth and row-level tenancy, versioned object storage, n8n parent workflow with bounded named specialists, fixed fixtures/tests and a single controlled write path. Reuse components; do not create a decorative multi-agent swarm.

BUILD ORDER

1 inspect existing repository; 2 document smallest change plan; 3 data schema/migrations; 4 source intake/versioning; 5 extraction + trace; 6 checks/conflicts; 7 batch comparison; 8 draft + evidence contract; 9 human review/revision; 10 source-change dependency map; 11 validation/export; 12 UI polish; 13 adverse/security/recovery tests; 14 deploy controlled demo and verify endpoint.

USAGE OPTIMIZATION

Patch existing code rather than rewriting. Keep explanations short. Generate only necessary files/code. Reuse schemas, UI and workflow components. Cache deterministic results by source/config revision. Use AI only where it adds value; run deterministic validation first/after.

SUCCESS

A user can upload approved synthetic/public-safe CMC sources, see exact extracted values and coordinates, resolve gaps/conflicts through named human review, produce a fully source-backed draft, observe stale dependency reopening after source change, execute tests and export a reproducible evidence package with no known P0/P1 defects in the tested critical journey.

17. Final acceptance checklist

- Canonical feature IDs and meanings preserved.
- 12 shared layers reused; no unnecessary parallel architecture.
- Proposed enhancements clearly labeled as proposed until built/tested.
- Every material output preserves exact evidence/source/version linkage.
- Human authority is explicit at every regulated decision/closure/release boundary.
- Normal path and adverse path both demonstrated.
- Stale revision, missing evidence, conflict, wrong tenant/role and retry exhaustion are visible states.
- Website journeys match backend states; no fake buttons or decorative workflows in the controlled demo.
- Frontend, API, database, orchestration and workers share typed contracts and correlation IDs.
- Test evidence, screenshots, README/case study, limitations and deployment status are truthful.